Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Jordan Green	Email	tipa@jhu.edu
Institute Name	Johns Hopkins University			
Address	JHU CMSC South, Suite 6122, Box 6-1 600 N. Wolfe St. Baltimore, MD 21287			

2. Technology Overview

Title of Proposed Technology	Once-Weekly Oral GLP-1 Receptor Agonist Delivery Platform for the Treatment of Alcohol and Opioid Use Disorders			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Drug Delivery Systems	Industrial Biotechnology	Bio-Chemical Materials
Concept of Proposed Technology	The proposed technology is a **long-acting, once-weekly (QW) oral peptide delivery platform** designed to enable sustained systemic exposure of a **GLP-1 receptor agonist (GLP-1RA)** for the treatment of substance use disorders, with a primary focus on **Alcohol Use Disorder (AUD)** and exploratory application to **Opioid Use Disorder (OUD)** in the fentanyl era. GLP-1RAs have demonstrated pharmacological plausibility for reducing craving and maladaptive reward-driven behaviors, supported by emerging human evidence in AUD and signal-level evidence relevant to OUD outcomes. However, currently approved oral GLP-1RAs exhibit very low absolute bioavailability and require strict daily administration conditions, limiting adherence and real-world effectiveness—particularly in populations with substance use disorders. This project aims to overcome these limitations through the development of a **novel oral formulation capable of delivering sustained GLP-1RA exposure for seven days (0–168 hours)** following a single oral dose. By leveraging the intrinsic long elimination half-life of GLP-			

	1RAs and combining controlled release with absorption-enhancing strategies, the platform seeks to achieve reliable weekly coverage with minimized inter- and intra-subject variability. A QW oral regimen is expected to reduce treatment burden, avoid injection-associated triggers, improve adherence, and support broader population-level access to addiction treatment.
Rationale for Joint R&D Project	**Public Health and Clinical Rationale** Severe substance use disorders represent an escalating global public health crisis, with synthetic opioids such as fentanyl driving unprecedented overdose mortality and social disruption. Despite the availability of effective medications for AUD and OUD, substantial unmet needs remain in terms of accessibility, adherence, long-term retention, and treatment models adapted to the clinical complexity of the fentanyl era. A QW oral therapy offers a practical opportunity to reduce daily decision burden, improve persistence, and expand treatment reach—particularly for individuals who are disengaged from existing care pathways or reluctant to use injectable medications. **Rationale for Korea–Research Institute Collaboration** Successful execution of this program requires **simultaneous excellence in two domains**: 1. **Advanced oral peptide formulation and CMC development**, including release-rate control, absorption enhancement, variability reduction, and GMP-ready manufacturing; and 2. **Addiction-focused translational and clinical development**, including proof-of-concept study design, endpoint selection, and ethical execution of substance-use research. Because experimental studies involving controlled substances are highly restricted in Korea, early translational and clinical PoC studies must be conducted through research institutes with established regulatory approvals, infrastructure, and expertise in addiction research. This joint structure integrates **Korean SME leadership in formulation technology, CMC execution, and manufacturing** with **research institute leadership in addiction-specific translational science and clinical strategy**, ensuring scientific rigor, regulatory credibility, and accelerated development.
Key Words	1. Once-weekly oral GLP-1RA (long-acting oral peptide delivery) 2. Opioid Use Disorder (OUD) / fentanyl-era risk (exploratory clinical validation) 3. Oral GLP-1RA bioavailability optimization

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	**Overall Objective** To develop and validate a **once-weekly (QW) oral GLP-1RA delivery platform** and generate an **IND/CTA-ready development package** enabling efficient clinical progression for the treatment of AUD and exploratory evaluation in OUD. **Specific Objectives** • Define a Target Product Profile (TPP) for a once-weekly oral therapy suitable for community-based addiction care • Develop an oral GLP-1RA formulation capable of sustaining therapeutic systemic exposure for at least 24 hours in small animal (equivalent to seven days in human) • Mitigate oral absorption limitations and administration-condition sensitivity • Establish translational evidence linking exposure to clinically relevant addiction-related endpoints • Advance IND-enabling readiness, including CMC, nonclinical justification, and clinical protocol development • Align the development strategy with expedited regulatory pathways for serious conditions where applicable			
R&D Target Specifications	Key performance targets include: • **Dosing**: Single oral dose per day (Equivalent to QW dose in human) • **Exposure**: The human target of sustained systemic exposure from 0–168 hours with meaningful trough at 168 hours corresponds to approximately 0-24 hours in small animal based on allometric time scaling • **Oral Bioavailability**: ≥ five-fold improvement over existing oral GLP-1RA formulations or achievement of predefined therapeutic exposure targets • **PK Variability**: $AUC_{0–24}h$ coefficient of variation ≤ 50% in preclinical evaluation • **Alcohol Intake Reduction**: 20% reduction of alcohol intake • **Stability**: ≥ 6-month shelf life at room temperature • **Manufacturability**: Establish pilot-scale processes and define a clear pathway toward GMP manufacturing, with preliminary design-to-cost considerations.			
R&D Plan	The R&D program follows a **milestone-driven, end-to-end development pathway** progressing from formulation feasibility to IND-ready status. **Phase 1: Program Setup and TPP Definition (Months 0–3)** • Finalize TPP, target exposure profile, and safety boundaries • Consolidate translational and clinical evidence and define Go/No-Go criteria **Phase 2: Oral Formulation Feasibility and Screening (Months 1–10)** • Develop and evaluate multiple formulation concepts • Conduct in vitro dissolution, stress testing, and early PK feasibility studies • Down-select lead formulation prototypes			

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	Phase 3: Translational Endpoint and Biomarker Development (Months 2–10) • Define AUD primary and OUD exploratory endpoints • Align sampling schedules with weekly exposure dynamics **Phase 4: Lead Optimization and Bridging Strategy (Months 7–18)** • Optimize release, absorption, and stability performance • Define dissolution–exposure bridging criteria and administration instructions **Phase 5: Nonclinical PoC and Safety Bridging (Months 4–15)** • Conduct behavioral and mechanistic PoC studies • Evaluate formulation-specific safety risks **Phase 6: Scale-Up and Quality System Development (Months 10–26)** • Evaluate scalability of manufacturing processes and analytical methods at pilot scale • Initiate stability studies and establish preliminary control and quality strategies **Phase 7: Regulatory Alignment and IND/CTA-Ready Package (Months 18–36)** • Evaluate scalability of manufacturing processes and analytical methods at pilot scale • Initiate stability studies and establish preliminary control and quality strategies

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will lead **clinical and translational strategy in addiction**, including: • Definition of clinical hypotheses and PoC objectives for AUD and OUD • Selection and operationalization of clinically meaningful endpoints • Development of translational biomarker and sampling frameworks • Design and support of early clinical trial protocols • Oversight of ethical and regulatory compliance for substance-use research
Expected R&R of Korean SME Partner	The Korean SME partner will serve as **consortium lead and development owner**, including: • Overall program management, budgeting, and TIPA reporting • Development of the QW oral GLP-1RA formulation platform • Evaluate analytical methods at pilot scale • Conduct pilot-scale CMC studies and identify key requirements for scale-up and GMP manufacturing • Engage with regulatory frameworks and define an IND/CTA-enabling development pathway • Draft key regulatory documents and outline submission-ready modules

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Scientific and Technical Impact** • Validation of a QW oral GLP-1RA delivery platform with controlled exposure and variability • Generation of translational evidence linking pharmacokinetics to addiction-related outcomes • Creation of extendable IP and know-how applicable to other peptide-based CNS indications **Regulatory and Development Impact** • Establishment of an IND/CTA-ready data framework to support regulatory submissions • Generation of early clinical and preclinical evidence supporting the feasibility and safety of QW oral exposure **Public Health and Social Impact** • Expanded access to addiction treatment through a non-injectable, adherence-friendly therapy • Potential reduction in stigma and improvement in treatment persistence • Contribution to mitigation of the global addiction crisis, particularly in the fentanyl era **Industrial and Economic Impact** • Strengthening Korea's capabilities in oral peptide delivery and long-acting formulations • Creation of export-ready technologies and SME-led licensing opportunities • Establishment of a sustainable Korea–research institute collaboration model for high-impact therapeutics

6. References

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(3) Wang W, Volkow ND, Wang Q, et al. **Semaglutide and Opioid Overdose Risk in Patients With Type 2 Diabetes and Opioid Use Disorder**. *JAMA Netw Open*. 2024;7(9):e2435247.

(4) U.S. Food and Drug Administration (FDA). **Fast Track, Breakthrough Therapy, Accelerated Approval, Priority Review**. Content current as of June 12, 2023.

(5) U.S. Food and Drug Administration (FDA). **RYBELSUS (Semaglutide) tablets, for oral use — Prescribing Information**. Revised 02/2025.

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(9) Bækdal TA, Breitschaft A, Donsmark M, et al. **Effect of Various Dosing Conditions on the Pharmacokinetics of Oral Semaglutide, a Human Glucagon-Like Peptide-1 Analogue in a Tablet Formulation**. *Diabetes Ther*. 2021;12(7):1915-1927. doi:10.1007/s13300-021-01078-y